

Market Dynamics as a Learned Stochastic Landscape

A Principled Framework for Portfolio Optimisation on NSE/BSE

market-agent project

September 2026

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1 Problem Framing

We treat the equity market as a **time-dependent dynamic landscape**: a high-dimensional stochastic system whose state at any moment is a composite of price history, news events, macroeconomic indicators, financial filings, and other observable signals. Our portfolio is a *low-dimensional cross-section* of this landscape — a projection onto a small number of assets — and we seek to find the projection that maximises the growth of invested capital over time.

Crucially, we do not have a first-principles equation of motion for the market. The dynamics must be *learnt from data*.

This framing leads naturally to **stochastic optimal control on a partially observed, data-driven dynamical system**.

2 Mathematical Formulation

2.1 Market State

Let $\mathbf{m}_t \in \mathbb{R}^D$ be the market state at discrete time step t (one trading day). It is a composite vector:

$$\mathbf{m}_t = \left[\underbrace{\mathbf{p}_t}_{\text{prices}}, \underbrace{\mathbf{n}_t}_{\text{news embeddings}}, \underbrace{\mathbf{f}_t}_{\text{macro / fundamentals}}, \underbrace{\mathbf{s}_t}_{\text{sentiment}} \right]$$

The true dynamics are unknown but assumed to be a first-order Markov process (after sufficient history is included in the state):

$$\mathbf{m}_{t+1} = F_\theta(\mathbf{m}_t) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(\mathbf{0}, \Sigma_\epsilon) \quad (1)$$

where F_θ is a neural network to be learnt from historical data.

2.2 Portfolio as a Cross-Sectional Projection

Let $\mathbf{S}_t \in \mathbb{R}^N$ be the vector of prices for N stocks. A portfolio is a weight vector $\mathbf{w}_t \in \mathbb{R}^N$ with $\sum_i w_{i,t} = 1$, $w_{i,t} \geq 0$ (long-only). Portfolio value evolves as:

$$V_{t+1} = V_t \cdot (1 + \mathbf{w}_t^\top \mathbf{r}_{t+1}),$$

where $\mathbf{r}_{t+1} \in \mathbb{R}^N$ is the vector of simple returns: $r_{i,t+1} = (S_{i,t+1} - S_{i,t})/S_{i,t}$.

The portfolio is a *linear projection* of the market onto a simplex; the “low-dimensional cross-section” constraint means $N \ll D$ and we cap concentration $w_i \leq w_{\max}$.

2.3 Objective: Growth-Optimal (Kelly) Criterion

We maximise *expected log-growth* over a horizon T :

$$\max_{\{\mathbf{w}_t\}_{t=1}^T} \mathbb{E} \left[\sum_{t=1}^T \log(1 + \mathbf{w}_t^\top \mathbf{r}_{t+1}) \right] \quad (2)$$

subject to $\mathbf{w}_t \in \Delta^{N-1}$ and market constraints.

Remark 1 (Why Kelly, not Sharpe). *The Sharpe ratio is not a Markovian reward — it depends on the full history of returns. Log-growth is additive and time-separable, making it a proper reward signal for sequential decision-making. Kelly-optimal portfolios also provably maximise long-run wealth almost surely.*

2.4 The Key Insight: Distribution, not Point Prediction

Maximising (2) does *not* require predicting individual returns $r_{i,t+1}$ precisely. It requires estimating the conditional distribution:

$$p(\mathbf{r}_{t+1} \mid \mathbf{m}_t)$$

well enough to compute the optimal weights $\mathbf{w}_t^*$. Under a conditional Gaussian approximation with mean $\hat{\boldsymbol{\mu}}_t$ and covariance $\hat{\Sigma}_t$, the Kelly objective reduces to a **mean-variance quadratic programme**:

$$\mathbf{w}_t^* = \arg \max_{\mathbf{w} \in \Delta} \hat{\boldsymbol{\mu}}_t^\top \mathbf{w} - \lambda \mathbf{w}^\top \hat{\Sigma}_t \mathbf{w} - \kappa \|\mathbf{w} - \mathbf{w}_{t-1}\|_1 \quad (3)$$

where:

- $\lambda > 0$ — risk-aversion parameter,
- $\kappa > 0$ — turnover penalty encoding transaction costs ($\approx 0.5\%$ roundtrip on NSE including brokerage, STT, stamp, GST),
- $\|\mathbf{w} - \mathbf{w}_{t-1}\|_1$ — ℓ_1 turnover; convex, differentiable almost everywhere.

This QP is solvable in closed form for small N , or via projected gradient descent. Crucially, the QP is **differentiable with respect to its inputs** ($\hat{\boldsymbol{\mu}}_t, \hat{\Sigma}_t$), so gradients flow back into the encoder.

3 Architecture

The system has three coupled, jointly trainable components.

3.1 Multi-Modal Market Encoder

Encodes the heterogeneous market state $\mathbf{m}_t$ into a compact latent representation $\mathbf{z}_t \in \mathbb{R}^{128}$:

$$\begin{aligned} \mathbf{e}_{\text{price}} &= \text{MarketCNN}(\text{OHLCV}_{t-28:t}, \text{vol}_{t-28:t}) \in \mathbb{R}^{64} \\ \mathbf{e}_{\text{news}} &= \text{NewsEncoder}(\text{LLM embeddings}_{t-5:t}) \in \mathbb{R}^{32} \\ \mathbf{e}_{\text{macro}} &= \text{MLP}(\text{rates}, \text{VIX-equiv}, \text{sector indices}) \in \mathbb{R}^{16} \\ \mathbf{e}_{\text{port}} &= \text{MLP}(\mathbf{w}_{t-1}, \text{P\&L}, \text{days held}) \in \mathbb{R}^{16} \\ \mathbf{z}_t &= \text{CrossAttention}([\mathbf{e}_{\text{price}}; \mathbf{e}_{\text{news}}; \mathbf{e}_{\text{macro}}; \mathbf{e}_{\text{port}}]) \in \mathbb{R}^{128} \end{aligned}$$

The cross-attention fusion allows each modality to attend selectively to the others — e.g. price patterns are weighted more heavily when no recent news exists.

3.2 World Model (Latent Dynamics)

A **Recurrent State Space Model** (RSSM) operating in the latent space:

$$\hat{\mathbf{z}}_{t+1} = \underbrace{d_\phi(\mathbf{h}_t)}_{\text{deterministic backbone}} + \underbrace{\sigma_\phi(\mathbf{h}_t) \cdot \boldsymbol{\eta}}_{\text{stochastic residual}}, \quad \boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, I) \quad (4)$$

where $\mathbf{h}_t = \text{GRU}(\mathbf{z}_t, \mathbf{h}_{t-1})$ is a deterministic hidden state.

The stochastic residual $\sigma_\phi(\mathbf{h}_t)$ encodes epistemic uncertainty: large in volatile or news-heavy regimes, small in trending ones. The world model enables **multi-step latent rollouts** without re-querying real data — key for planning and for backpropagating through time.

3.3 Conditional Return Estimator

Two lightweight heads on $\mathbf{z}_t$:

$$\hat{\boldsymbol{\mu}}_t = \mu_\psi(\mathbf{z}_t) \in \mathbb{R}^N \quad (\text{expected 1-week returns}) \quad (5)$$

$$\hat{L}_t = L_\psi(\mathbf{z}_t) \in \mathbb{R}^{N \times K} \quad (\text{Cholesky factor, } K \ll N) \quad (6)$$

$$\hat{\Sigma}_t = \hat{L}_t \hat{L}_t^\top + \text{diag}(\hat{\mathbf{d}}_t) \quad (\text{low-rank + diagonal covariance}) \quad (7)$$

The low-rank structure ($K \approx 10\text{--}20$ latent factors) keeps the covariance estimate tractable and imposes a factor model structure that mirrors how markets actually move (a few common factors driving most co-movement).

3.4 Differentiable Portfolio Optimiser

The QP in (3) is solved at each time step. It takes $(\hat{\boldsymbol{\mu}}_t, \hat{\Sigma}_t, \mathbf{w}_{t-1})$ as inputs and returns the optimal weights $\mathbf{w}_t^*$. NSE-specific hard constraints are layered on:

- **Simplex:** $\mathbf{w} \geq \mathbf{0}$, $\mathbf{1}^\top \mathbf{w} = 1$
- **Concentration cap:** $w_i \leq 0.15$
- **T+1 settlement:** w_i cannot decrease on the day after purchase (stocks bought at t are locked until $t+1$)
- **Circuit filters:** zero out weights for stocks on upper/lower circuit

4 End-to-End Training

The full stack is differentiable:

$$\underbrace{\mathbf{m}_t}_{\text{observations}} \xrightarrow{\text{Encoder}} \mathbf{z}_t \xrightarrow{\text{Return heads}} (\hat{\boldsymbol{\mu}}_t, \hat{\Sigma}_t) \xrightarrow{\text{QP}} \mathbf{w}_t^* \xrightarrow{\text{market}} \underbrace{\log(1 + \mathbf{w}_t^{*\top} \mathbf{r}_{t+1})}_{\text{reward}}$$

Gradients of the portfolio return flow back through the QP (via the KKT conditions of the active constraints) and into the encoder and return heads.

4.1 Composite Training Loss

$$\mathcal{L} = \underbrace{-\sum_t \log(1 + \mathbf{w}_t^{*\top} \mathbf{r}_{t+1})}_{\mathcal{L}_{\text{Kelly}}} + \underbrace{\alpha \sum_t \|\hat{\mathbf{z}}_{t+1} - \mathbf{z}_{t+1}\|^2}_{\mathcal{L}_{\text{world model}}} + \underbrace{\beta \sum_t \|\hat{\boldsymbol{\mu}}_t - \mathbf{r}_{t+1}\|^2}_{\mathcal{L}_{\text{return}}} + \underbrace{\gamma \Omega(\mathbf{w}_t^*)}_{\mathcal{L}_{\text{reg}}} \quad (8)$$

- $\mathcal{L}_{\text{Kelly}}$ — primary objective: maximise log-growth
- $\mathcal{L}_{\text{world model}}$ — next-state prediction; anchors the latent representation to observable market dynamics
- $\mathcal{L}_{\text{return}}$ — return calibration; ensures $\hat{\boldsymbol{\mu}}_t$ is informative even when $\mathcal{L}_{\text{Kelly}}$ gradients are noisy
- $\mathcal{L}_{\text{reg}}$ — concentration and drawdown regularisation

4.2 Training Phases

Algorithm 1 Staged Training Protocol

- 1: **Phase 1 — World Model Pretraining** (self-supervised)
 - 2: Mask 20% of $\mathbf{m}_t$ at random; train Encoder + RSSM to reconstruct.
 - 3: Train RSSM on sequences: given $\mathbf{z}_1, \dots, \mathbf{z}_T$, predict $\mathbf{z}_{T+1}$.
 - 4: Loss: $\mathcal{L}_{\text{world model}}$ only. Freeze after convergence.

 - 5: **Phase 2 — Return Distribution Calibration**
 - 6: Fix Encoder + RSSM. Train return heads (μ_ψ, L_ψ) .
 - 7: Loss: Gaussian log-likelihood of observed $\mathbf{r}_{t+1}$.
 - 8: Validate: IC (Spearman corr between $\hat{\mu}_{i,t}$ and $r_{i,t+1}$) > 0.03 .

 - 9: **Phase 3 — End-to-End Portfolio Training**
 - 10: Unfreeze all components. Optimise (8) with the full composite loss.
 - 11: Use latent rollouts (via RSSM) for multi-step policy improvement.

 - 12: **Phase 4 — Online Adaptation**
 - 13: Deploy with small learning rate $\eta_{\text{online}} \ll \eta_{\text{train}}$.
 - 14: Regime detection: if $\log p(\mathbf{z}_t | \mathbf{z}_{t-1}) \ll \bar{\ell}$, reduce all w_i until re-calibration.
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5 Why This Avoids the Zero-Predictor Failure

The previous CNN was trained with an MSE loss on individual log-returns:

$$\mathcal{L}_{\text{CNN}} = \sum_{i,t} (\hat{r}_{i,t} - r_{i,t})^2$$

The global minimum of this loss is $\hat{r}_{i,t} = 0 \forall i, t$ because returns are approximately zero-mean. The model learnt nothing.

Under the Kelly objective (2), the loss at the *equal-weight portfolio* baseline (the analogue of “predict zero”) is:

$$\mathcal{L}_{\text{Kelly}}|_{\mathbf{w}=\frac{1}{N}\mathbf{1}} = -\sum_t \log \left(1 + \frac{1}{N} \sum_i r_{i,t+1} \right) \approx -(\text{market return})$$

Any model that learns to overweight the top-quartile stocks versus the bottom quartile obtains a strictly better objective value, because cross-sectional variance of returns is large ($\approx 2\text{--}4\%$ weekly standard deviation between the best and worst deciles on NSE). **The cross-sectional signal exists even when aggregate direction is unpredictable** — market returns cancel in the relative comparison.

Remark 2 (Information Coefficient as diagnostic). *During Phase 2, before committing to end-to-end training, validate signal quality via the Information Coefficient (IC):*

$$IC_t = \text{Spearman}(\hat{\mu}_{i,t}, r_{i,t+1})$$

A time-average IC > 0.03 is tradeable, > 0.05 is usable, > 0.10 is excellent. If IC is near zero after Phase 2, the encoder has not learnt any cross-sectional signal — stop before Phase 3 and investigate features.

6 The Koopman Perspective

The encoder implicitly learns a **Koopman embedding**: a lifted feature space in which the nonlinear market dynamics become approximately linear.

Definition 1 (Koopman operator). *For a dynamical system $\mathbf{m}_{t+1} = F(\mathbf{m}_t)$, the Koopman operator $\mathcal{K}$ acts on observables $g : \mathcal{M} \rightarrow \mathbb{R}$ as:*

$$(\mathcal{K}g)(\mathbf{m}) = g(F(\mathbf{m}))$$

$\mathcal{K}$ is linear and infinite-dimensional, but can be approximated by a finite-dimensional matrix in a learnt basis.

Classical factor models (Fama-French, etc.) are first-order linear Koopman approximations in a PCA basis. Our encoder learns *nonlinear, data-driven Koopman eigenfunctions* — the directions in which market dynamics are most predictable — and the RSSM transition operates linearly in this latent space.

This also explains why the low-rank covariance structure is natural: in the Koopman basis, K dominant eigenmodes drive most of the cross-sectional co-movement, with idiosyncratic residuals treated as diagonal noise.

7 NSE-Specific Constraints and Market Microstructure

Constraint	Mechanism	Modelling
T+1 settlement	Bought stock locked 1 day	Hard QP constraint
Circuit limits	$\pm 10/20\%$ price bands	Zero out w_i pre-QP
Concentration cap	Single position $\leq 15\%$	QP box constraint
Transaction costs	$\approx 0.5\%$ roundtrip	ℓ_1 turnover penalty
Illiquidity	Cost $\propto \Delta w_i / \text{ADV}_i$	Weighted turnover penalty
No shorting	Long-only	$w_i \geq 0$ simplex

Table 1: NSE market constraints and their implementation in the optimiser.

The illiquidity-adjusted turnover penalty replaces the uniform κ in (3):

$$\mathcal{L}_{\text{turnover}} = \sum_i \kappa_i |w_{i,t} - w_{i,t-1}|, \quad \kappa_i \propto \frac{1}{\text{ADV}_i}$$

where ADV_i is the 20-day average daily traded value of stock i . This automatically discourages large moves in illiquid names.

8 Data Sources and Feature Mapping

LLM-extracted signals (via `llm_extract.jl` and `NewsMonitor`) produce scalar features per company per event:

$$\mathbf{n}_t = [\text{sentiment}, \text{severity}, \text{revenue_surprise}, \text{guidance_direction}, \text{days_since_last_event}, \dots] \in \mathbb{R}^{15}$$

These become part of the encoder’s news modality $\mathbf{e}_{\text{news}}$.

Source	Signal extracted	Existing module
OHLCV (daily + hourly)	Price momentum, volatility, volume patterns	InferenceCache
BSE corporate filings	Results surprise, board decisions, shareholding	NewsMonitor (bse.jl)
Earnings call transcripts	Guidance revisions, management tone	TijoriData (fetch_document)
Financial news RSS	Sector catalysts, analyst upgrades	NewsMonitor (rss.jl)
ICRA/CRISIL ratings	Credit events, downgrade/upgrade	(to build)
RBI policy releases	Rate-sensitive sector impact	(to build)
SIAM/AMFI monthly data	Auto/fund-flow sector signals	(to build)

Table 2: Information sources feeding the multi-modal encoder.

9 Build Order

The system decomposes into four increasingly complex milestones:

- 1. Cross-sectional return estimator + Markowitz QP**
Input: OHLCV only ($\mathbf{e}_{\text{price}}$). No world model yet. Estimate $(\hat{\boldsymbol{\mu}}_t, \hat{\Sigma}_t)$ and solve the QP. Validate via IC. This is the minimum viable portfolio model.
- 2. Add the world model (RSSM)**
Pretrain on price sequences. Enables multi-step latent rollouts and sharper gradient signal during Phase 3.
- 3. Add multi-modal features**
Incorporate LLM news embeddings, macro indicators. Measure IC lift from each modality independently.
- 4. End-to-end portfolio training**
Full composite loss (8). Differentiable QP layer. Online adaptation with regime detection.

Each milestone produces a *runnable system* with measurable IC and backtested returns — no dead ends where months of work yield nothing evaluable.